

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems

Experiment: 8 Steering Angle Variation in MIMO Beamforming Using MATLAB

Objective – 1 Analyze Steering Angle (Degrees)

Program:

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

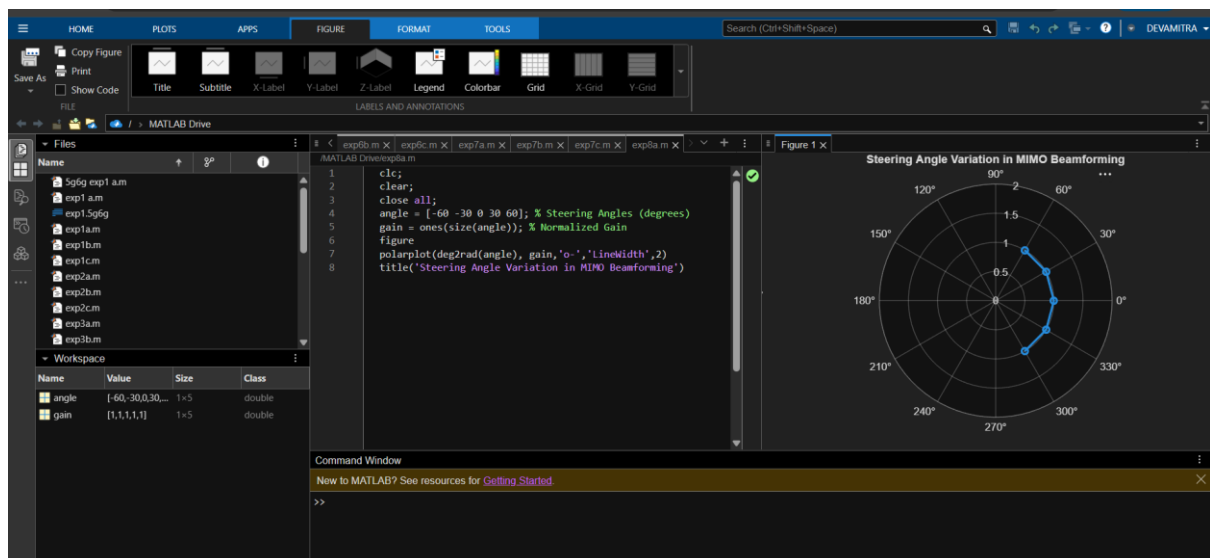
gain = ones(size(angle)); % Normalized Gain

figure

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```

Output:



Objective – 2 Compare Array Gain (dB)

Program:

```
clc;

clear;

close all;
```

```
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
```

```
gain = [9 12 15 12 9]; % Array Gain (dB)
```

```
figure
```

```
bar(angle, gain)
```

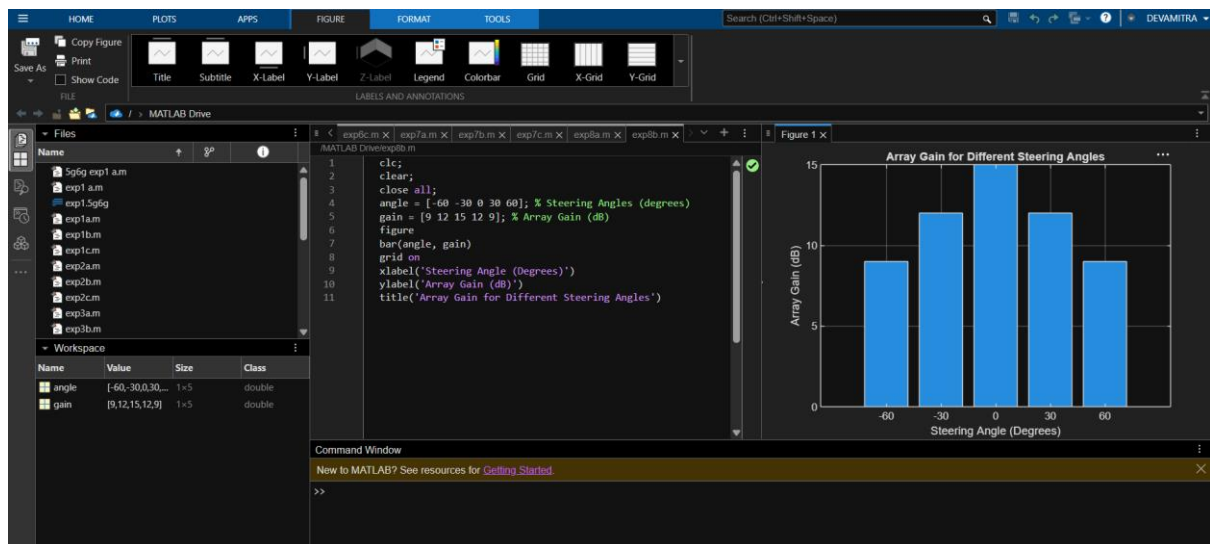
```
grid on
```

```
xlabel('Steering Angle (Degrees)')
```

```
ylabel('Array Gain (dB)')
```

```
title('Array Gain for Different Steering Angles')
```

Output:



Objective – 3 Evaluate Beam Direction Error (Degrees)

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees) actual = [-58 -32 1  
29 62]; % Actual Beam Direction (degrees)
```

```
error = abs(desired - actual); % Beam Direction Error
```

```
figure
```

```
plot(desired, error, 'o-', 'LineWidth', 2)
```

grid on

xlabel('Desired Steering Angle (Degrees)')

ylabel('Beam Direction Error (Degrees)')

title('Beam Direction Error for Different Steering Angles')

Output:

